Motion in a straight line

Tue May 05 2026



$S_{n^{th}}$ = Disp. in n^{th} second $\Rightarrow$ from $t = n-1$ till $t = n$ (disp.)

$S_{n^{th}} = S_n - S_{n-1}$

$S = ut + \frac{1}{2}at^2$

$= \left(un + \frac{1}{2}an^2\right) - \left[u(n-1) + \frac{1}{2}a(n-1)^2\right]$

$= \cancel{un} + \cancel{\frac{1}{2}an^2} - \cancel{un} + u - \cancel{\frac{1}{2}an^2} - \frac{1}{2}a + \cancel{2}n\frac{a}{\cancel{2}}$

$= u - \frac{a}{2} + na$

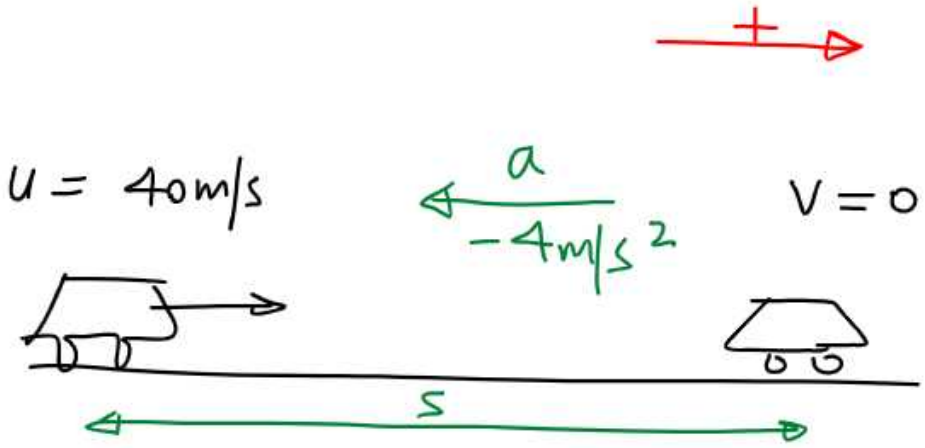
$= u + \frac{a}{2}\left[2n - 1\right]$

$u = 40\,m/s$

Q: Driver applies the brakes such that retardation is $4\,m/s^2$. Find

a) time in which car will stop

b) stopping distance



a) $v = u + at$

$0 = 40 - 4 \times t$

$t = 10\,sec$

b) $s = ut + \frac{1}{2}at^2$

$= 40 \times 10 + \frac{1}{2}(-4) \times 10 \times 10$

$= 400 - 200 = 200\,m$

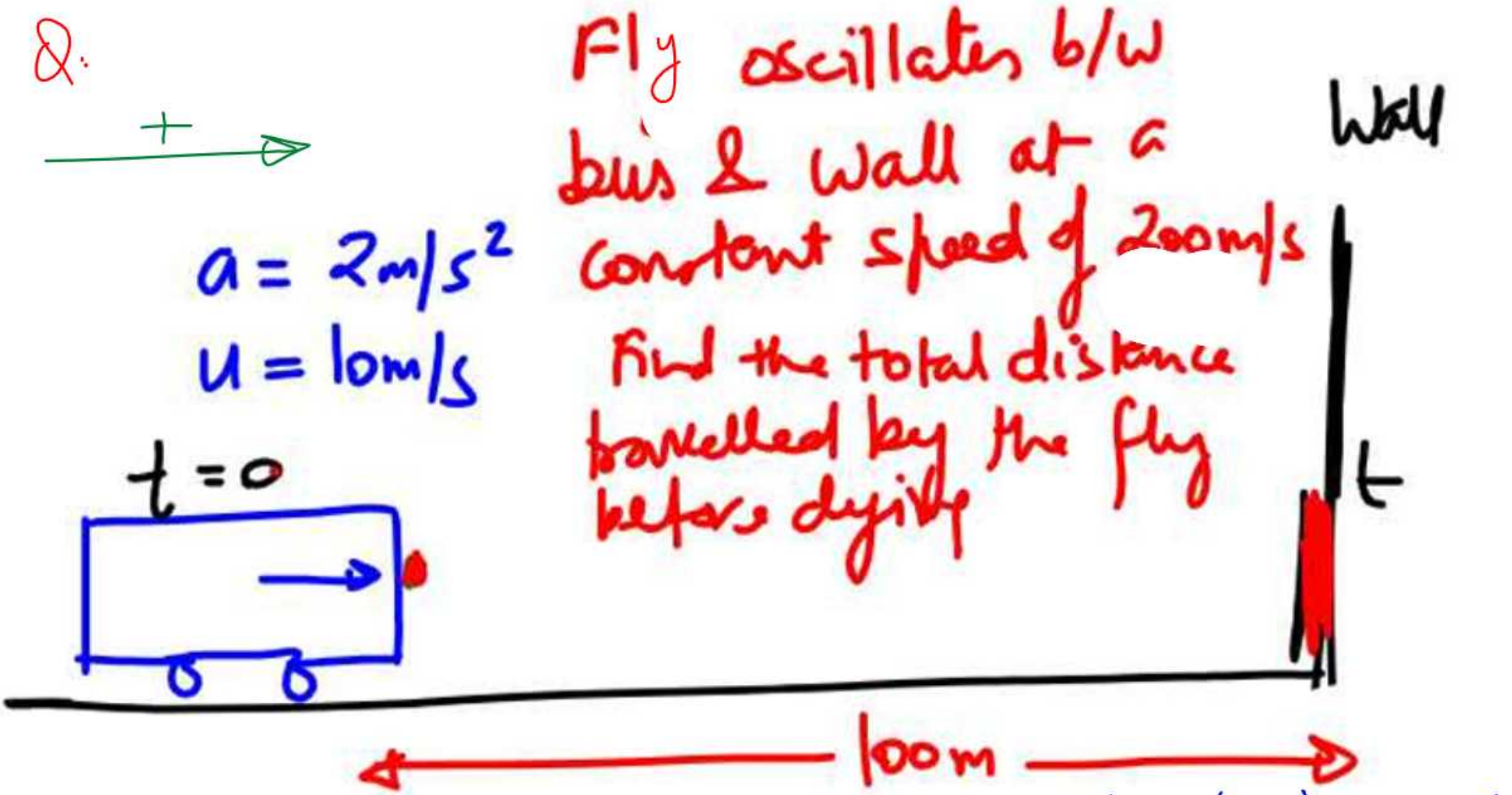
$v^2 = u^2 + 2as$

$0 = 40 \times 40 - 2 \times 4 \times s$

$8s = 40 \times 40$

$s = 200\,m$

Q.



$s = ut + \frac{1}{2}at^2$

$100 = 10 \times t + \frac{1}{2} \times 2 \times t^2$

$t^2 + 10t - 100 = 0$

$t = \frac{-10 \pm \sqrt{100 + 400}}{2}$

$= \frac{-10 + 10\sqrt{5}}{2}$

$t = \frac{10(\sqrt{5}-1)}{2} = 5(\sqrt{5}-1)$

$l = v \times t$

$= 200 \times 5(\sqrt{5}-1)$

$l = 1000(\sqrt{5}-1)$

$u=0$

A particle starts from rest with constant acceleration a. The ratio of distance covered in 3rd second to that in 1st second is:

(A) 3

(B) 5

(C) 7

(D) 9

$$S_{nm} = u + \frac{a}{2}(2n-1)$$

$$S_{1st} = 0 + \frac{a}{2}(2\times 1 - 1)$$

$$S_1 = \frac{1.a}{2}$$

$$S_{3rd} = 0 + \frac{a}{2}(5)$$

$$\frac{S_3}{S_1} = \frac{5a/2}{a/2} = 5:1$$

$$S_2 = 0 + \frac{a}{2}(4-1)$$

$$S_2 = \frac{3a}{2}$$

$$S_1 : S_2 : S_3 : S_4 = 1:3:5:7\ldots$$

A car accelerates uniformly from rest to 20 m/s in 10 s. Distance covered is:

(A) 50 m (B) 100 m (C) 200 m (D) 400 m

$u = 0$

$t = 10$

$v = 20$



$V = u + at$

$20 = 0 + a \times 10$

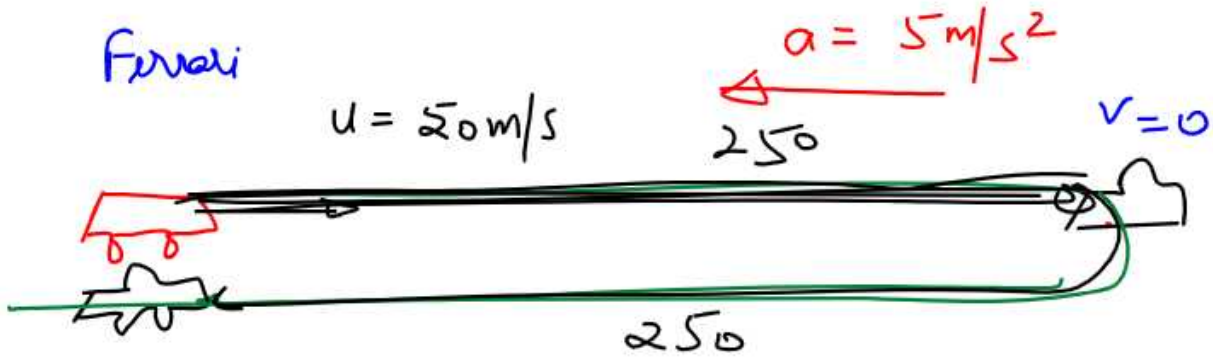
$a = 2 m/s^2$

$s = ut + \frac{1}{2}at^2$

$= 0 + \frac{1}{2} \times 2 \times 10 \times 10$

$= 100$

Q. Find the distance in first 20 seconds.



$$s = ut + \frac{1}{2}at^2$$

$$= 50 \times 20 + \frac{1}{2}(-5) \times 20 \times 20$$

$$= 1000 - 1000$$

$$s = 0$$

$$s = ut + \frac{1}{2}at^2$$

$$= 50 \times 10 - \frac{1}{2} \times 5 \times 10 \times 10$$

$$= 500 - 250$$

$$= 250$$

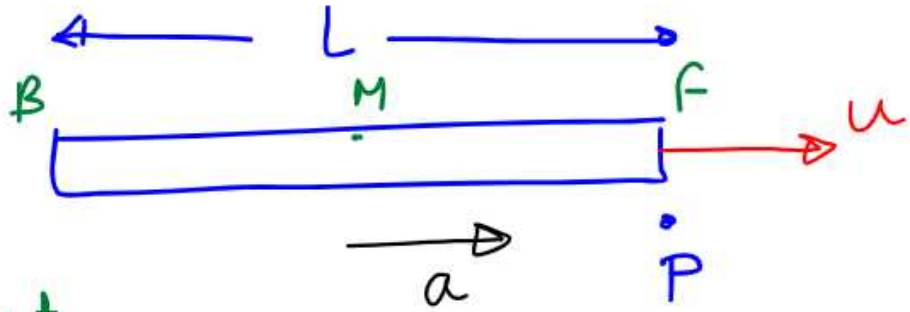
$$l = 2 \times 250 = 500$$

$$V = u + at$$

$$0 = 50 - 5t$$

$$t = 10 \text{ sec}$$

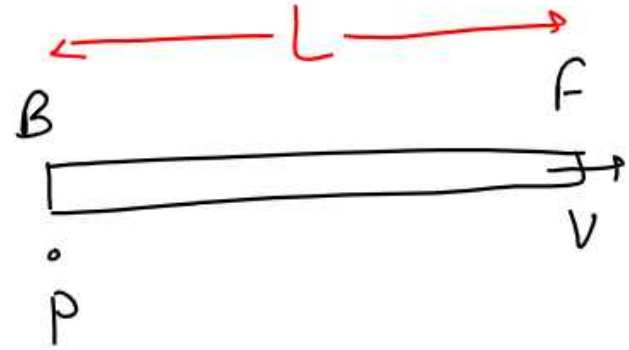
Q. F crosses P at u and B crosses P at v. Find the speed at which M will cross P

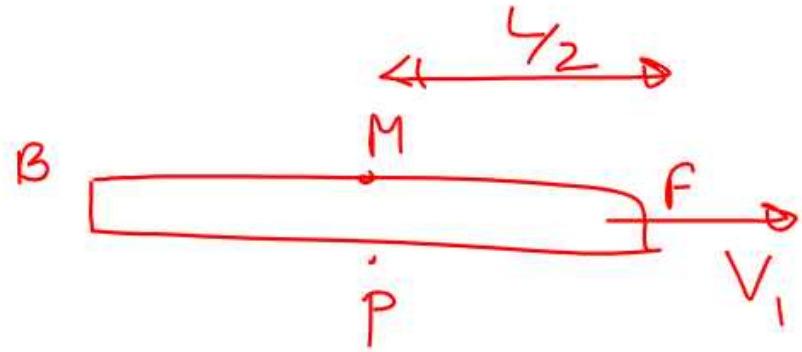


$v^2 = u^2 + 2as$

$v^2 = u^2 + 2 \times aL$

$v^2 - u^2 = 2aL$





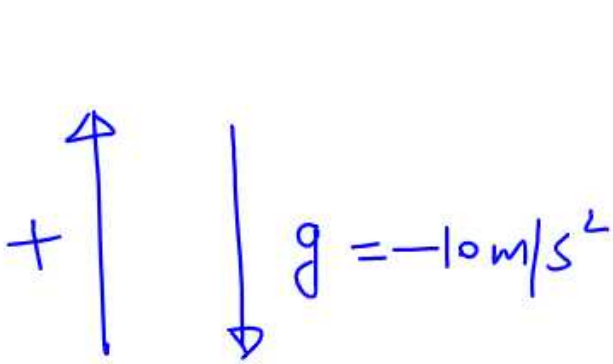
$V_1^2 = u^2 + 2a\frac{L}{2}$

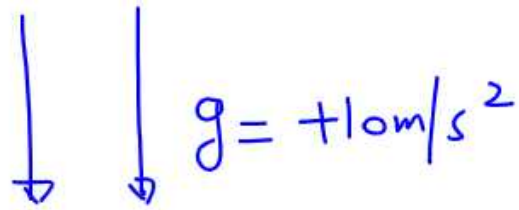
$V_1^2 - u^2 = aL = \frac{v^2 - u^2}{2}$ $\Rightarrow 2V_1^2 - 2u^2 = v^2 - u^2$

$\Rightarrow V_1^2 = \frac{v^2 + u^2}{2}$ $\Rightarrow V_1 = \sqrt{\frac{v^2+u^2}{2}}$

Vertical gravity

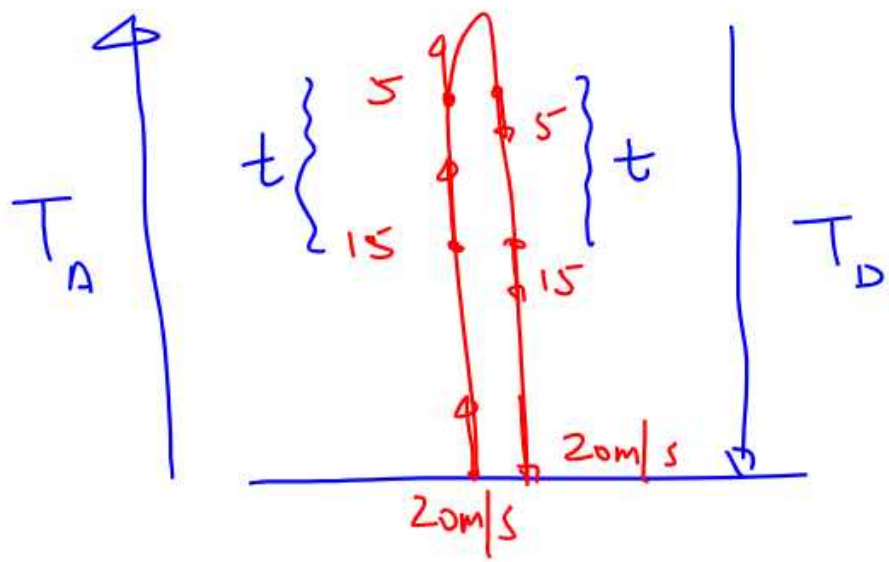






- To keep things simple, choose the direction of disp. as +

- Upward and downward motion are symmetric
  - $T_A = T_D = \frac{1}{2}T$



Q. Find the following

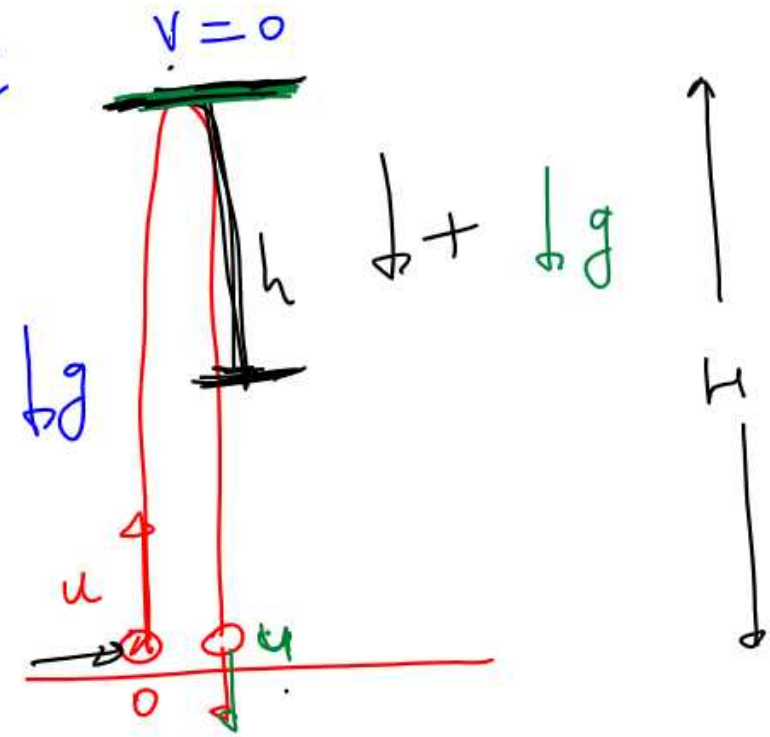
a) Time of Ascent → $T_A = u/g$

b) " " descent → $T_D = u/g$

$\Big\}\ T = 2u/g$

c) ~~Time~~ Time in falling through height h from top

d) Max height



a) $v = u + at$

$0 = +u - gt$

$t = u/g = T_A$

b) $T_D = u/g$

c) $s = ut + \frac{1}{2}at^2$

$h = 0 + \frac{1}{2}gt^2$

$\boxed{h = \frac{1}{2}gt^2}$

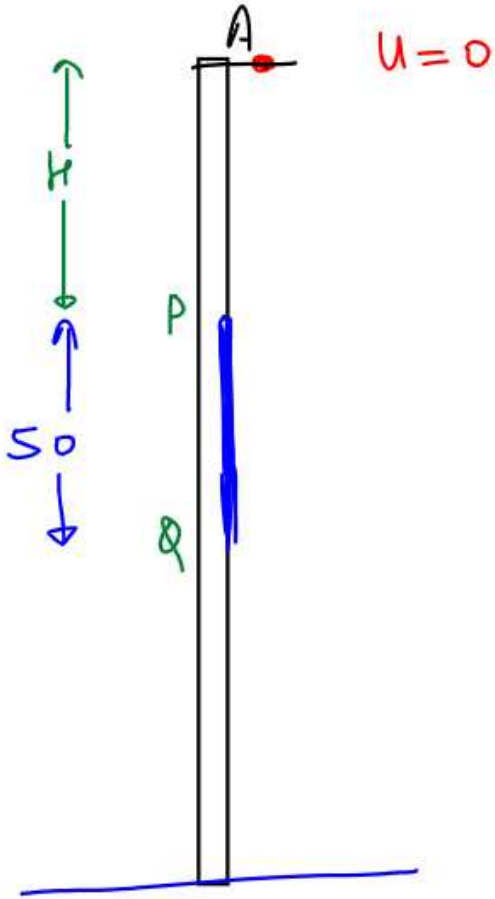
d) $H = \frac{1}{2}g \times \frac{u^2}{g^2} = \frac{u^2}{2g}$

$v^2 = u^2 + 2as$

$0 = u^2 - 2gH$

$H = u^2/2g$

Q. A particle dropped from top crosses the entire window in 2 seconds. If the length of window is 50m, then find the value of H.





$T_{PQ} = 2$ sec

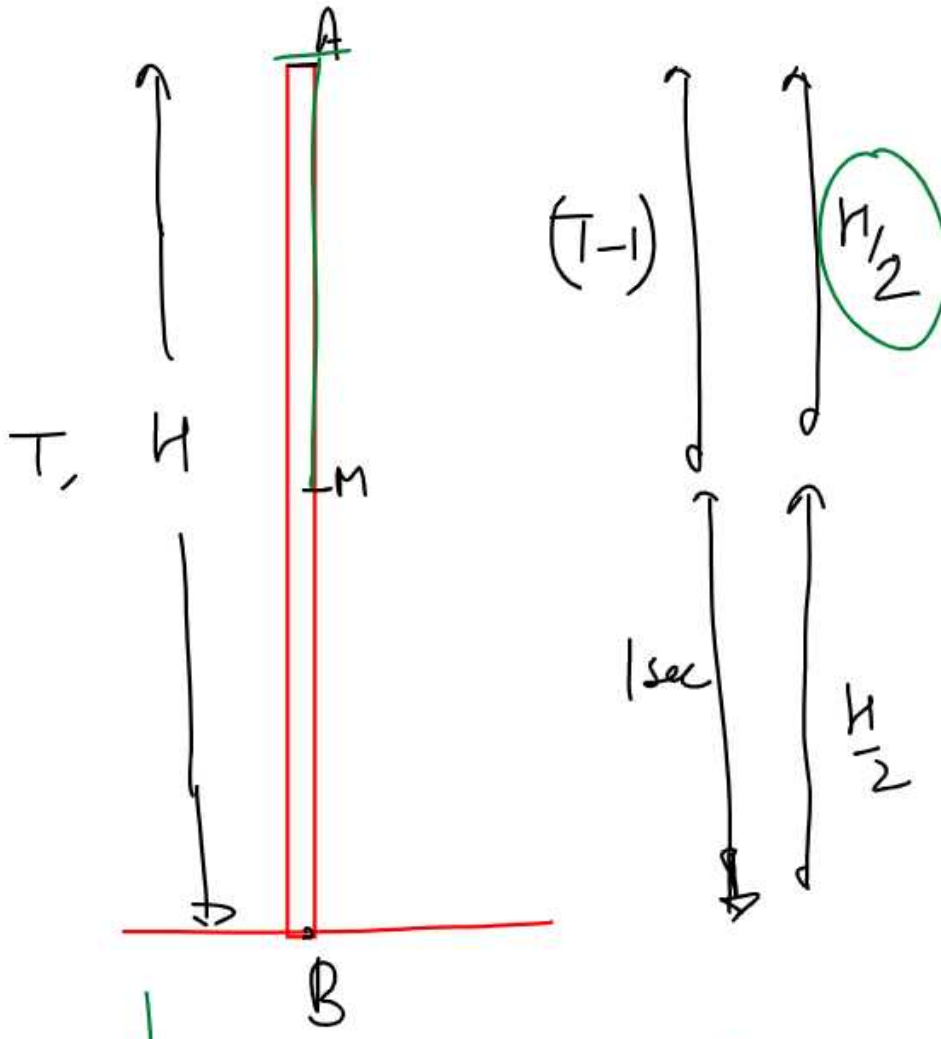
$H = \frac{1}{2} g T^2$ → $H = \frac{1}{2} \times 10 \times 1.5 \times 1.5 = \frac{22.5}{2} = 11.25$

$H + 50 = \frac{1}{2} g (T+2)^2 \Rightarrow \frac{1}{2} g T^2 + 50 = \frac{1}{2} g T^2 + \frac{1}{2} g 4 + \frac{1}{2} g 4T$

$50 = 20 + 20T$

$T = 1.5$ sec

Q. A ball dropped from the top of a tower travels half the distance in last second. Find the height of the tower.



$$H = \frac{1}{2} g T^2 \rightarrow (1)$$

$$\frac{H}{2} = \frac{1}{2} g (T-1)^2 \rightarrow (2)$$

(1) ÷ (2)

$$2 = \frac{T^2}{(T-1)^2}$$

$$\sqrt{2} = \frac{T}{T-1}$$

$$\sqrt{2}T - \sqrt{2} = T$$

$$\sqrt{2}T - T = \sqrt{2}$$

$$T = \frac{\sqrt{2}}{\sqrt{2}-1}$$

$$H = \frac{1}{2} g \left(\frac{\sqrt{2}}{\sqrt{2}-1}\right)^2$$

$$= 5 \left(\frac{\times 2}{(\sqrt{2}-1)^2}\right)$$

$$H = \frac{10}{(\sqrt{2}-1)^2}$$

Ball is dropped from height H_0.

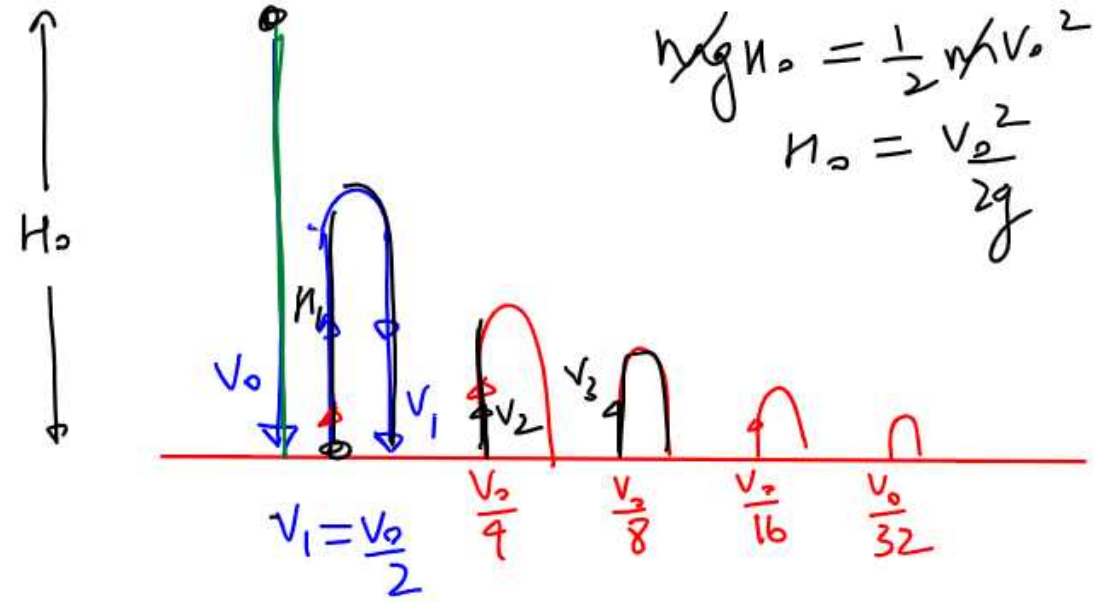
Q. After every collision, the speed of the ball becomes half

Find the following

a) Total distance travelled

b) " Time of Motion

$$H_0 = \frac{V_0^2}{2g}$$



$mgH_0 = \frac{1}{2}mV_0^2$

$H_0 = \frac{V_0^2}{2g}$

$$2H_1 = \frac{V_1^2}{2g} \times 2 = \frac{V_1^2}{g}$$

$$2H_2 = \frac{V_2^2}{2g} \times 2 = \frac{V_2^2}{g}$$

$$2H_3 = \frac{V_3^2}{2g} \times 2 = \frac{V_3^2}{g}$$

$$\vdots$$

$$l = H_0 + 2H_1 + 2H_2 + 2H_3 \ldots\ldots$$

$$l = \frac{V_0^2}{2g} + \frac{V_1^2}{g} + \frac{V_2^2}{g} + \frac{V_3^2}{g} \ldots\ldots$$

$$= \frac{V_0^2}{2g} + \left[\frac{V_0^2}{4g} + \frac{V_0^2}{16g} + \frac{V_0^2}{64g} + \frac{V_0^2}{256g}\right.$$

$$= \frac{V_0^2}{2g} + \frac{V_0^2}{4g}\left[1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} \ldots\ldots\right]$$

$$l = \frac{V_0^2}{2g} + \frac{V_0^2}{4g}\left[\frac{1}{1-1/4}\right]$$

$$= \frac{V_0^2}{2g} + \frac{V_0^2}{4g}\left[\frac{1}{3/4}\right]$$

$$= \frac{V_0^2}{2g} + \frac{V_0^2}{3g}$$

$$= \frac{V_0^2}{g}\left[\frac{1}{2} + \frac{1}{3}\right]$$

$$l = \frac{5}{6}\frac{V_0^2}{g}$$

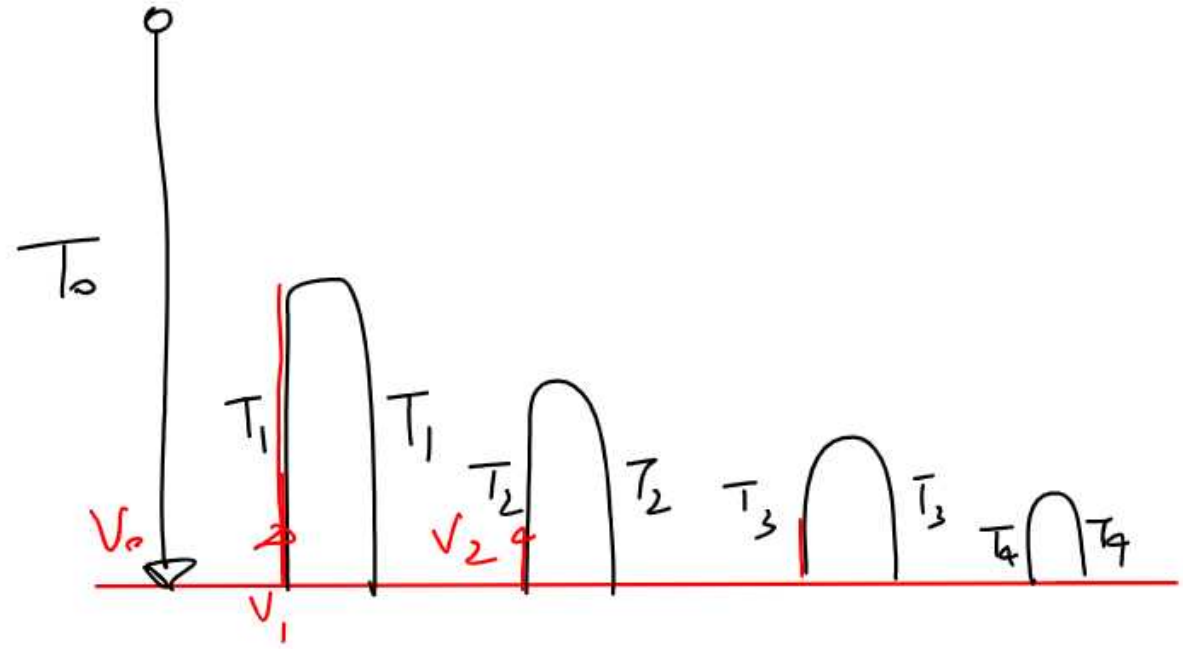
$$T_0 = \frac{V_0}{g}$$

$$T_1 = \frac{V_1}{g} = \frac{V_0}{2g}$$

$$T_2 = \frac{V_2}{g} = \frac{V_0}{4g}$$

$$T_3 = \frac{V_3}{g} = \frac{V_0}{8g}$$

$$T_4 = \frac{V_4}{g} = \frac{V_0}{16g}$$

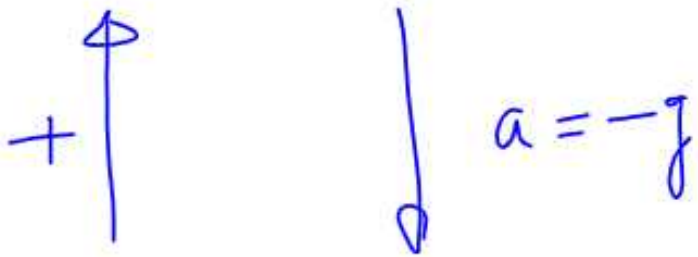


$$T = T_0 + 2T_1 + 2T_2 + 2T_3 + 2T_4 \cdots$$

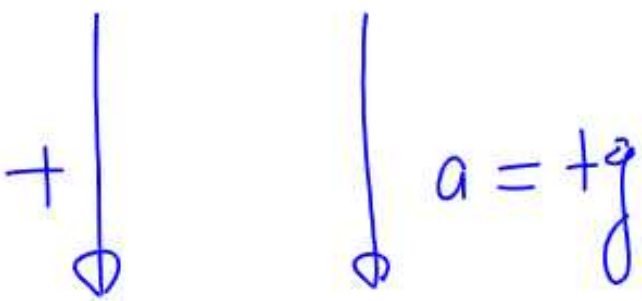
$$= \frac{V_0}{g} + \left[\frac{V_0}{g} + \frac{V_0}{2g} + \frac{V_0}{4g} + \frac{V_0}{8g} \cdots\right]$$

$$= \frac{V_0}{g} + \frac{V_0}{g}\left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \cdots\right]$$

$$= \frac{V_0}{g} + \frac{V_0}{g}\left[\frac{1}{1-1/2}\right] = \frac{V_0}{g} + \frac{2V_0}{g} = \frac{3V_0}{g}$$


+
a = −g




+
a = +g



Please Revise the session

by redoing the problems